

MODIFICATION OF THE DRUG RELEASE OF IBUPROFEN BY HOT-MELT COATING WITH MIXES OF COMPRITOL® 888 ATO AND NON-IONIC SURFACTANTS

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I - PURPOSE

Hot-Melt Coating with lipid excipients for taste-masking or stabilization often leads to a slow release of the drug. The addition of surfactants in lipid formulations has already shown an increase of the drug release for matrix tablets [1] and a stabilization of the lipid excipient [2]. That concept of modified drug release by adding surfactant in the lipid coating material was tested on Ibuprofen.

II - METHODS

• Mixes of lipid excipient and non-ionic surfactants

The lipid excipient used for Hot-Melt Coating is Compritol® 888 ATO (glyceryl behenate). The non-ionic surfactants used were PEG-8 behenate (surfactant moieties of Compritol® HD5), PEG-32 stearate (Stearate 1500), Polysorbate 60 and PEG-8 caprylic/capric glycerides (Labrasol®). They vary in term of hydrophilic-lipophilic balance (HLB), water-solubility, fatty acids chain length and physical state (i.e. solid or liquid). Each of these surfactants was mixed with Compritol® 888 in order to get a coating mix with an average HLB of 5.

• Thermal characterization of coating mixes

The influence of these surfactants on the thermal properties of Compritol® 888 was assessed by DSC (Pyris Diamond, Perkin Elmer, USA) and drop point measurement (FP83HT Dropping Point Cell and FP90 Central Processor, Mettler Toledo, Switzerland).

• Formulation of modified-release ibuprofen capsules and tablets

Ibuprofen (Albemarle, USA) was heated in a top-spray fluid bed coater (GPCG 1.1, Glatt, Germany). Compritol® 888 or the coating mix was melted with microwaves and then kept in its liquid form in a stirred-beaker. The coating agent or mix was then sprayed on ibuprofen at a concentration of 3% (w/w). Coated products were cooled in the fluid bed system to room temperature.

Coated products were blend with 50% (w/w) of microcrystalline cellulose (Avicel™ PH101, FMC, USA) in a Turbula blender (T2C, WAB, Switzerland) for 5 minutes at 62 rpm. These mixtures were then used either in size 00 capsules or for tableting (EK0, Korsch, Germany). Tablets present an apparent density of about 1.15 g/mL.

Dissolution profiles were evaluated by the dissolution test (Sotax, Switzerland) described in the USP monograph 'Ibuprofen tablets'.

III - RESULTS

- The four surfactants are partially soluble in Compritol® and decrease the lipid's drop point. The fluidization air temperature was decreased from 62°C for Compritol® to 58 or even 50°C in order to obtain a solid fraction of 70 to 80% during the crystallization of the mixture as observed by DSC [3].

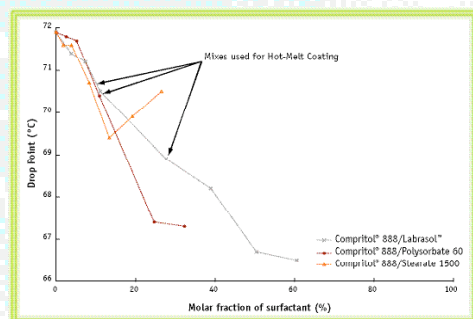


Figure 1: Compritol® 888 drop point variation as a function of the molar fraction of surfactant in the mix.

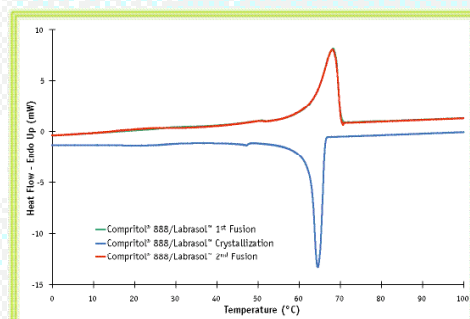


Figure 2: DSC of the coating mix Compritol® 888 + Labrasol® 80/20 w/w at 3°C/Min.

- All formulations in capsules present an immediate release of the drug. A quicker release can be noted for formulations containing water-soluble surfactants (Figure 3).

Dissolution efficiencies (DE) range from 71 to 89%.

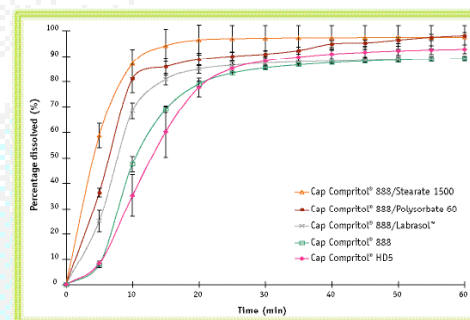


Figure 3: Dissolution profiles of ibuprofen coated with mix of Compritol® 888 and non-ionic surfactant - in capsules.

- After compaction, ibuprofen coated with Compritol® presents a slow release (DE = 17%) due to the formation of a lipid matrix and the reduced surface area between the dosage form and the dissolution medium. Figure 4 shows that all surfactants tested help to increase the dissolution efficiency of tablet formulations (from 33 to 72%).

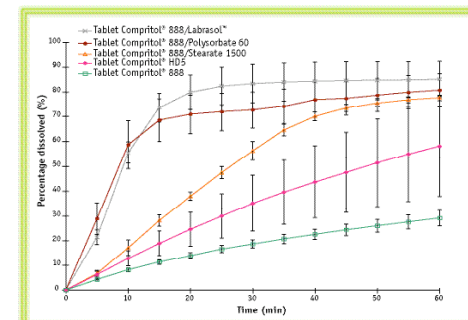


Figure 4: Dissolution profiles of ibuprofen coated with mix of Compritol® 888 and non-ionic surfactant - in tablets.

Table 1: Dissolution efficiencies of the five coated ibuprofen in capsule or tablet formulations (mean±sd).

Coating mix	DE of capsule formulations (%)	DE of tablet formulations (%)
Compritol® 888	71±1	17±2
Compritol® HD5	71±2	33±11
Compritol® 888 + Stearate 1500	89±5	49±1
Compritol® 888 + Polysorbate 60	83±2	67±7
Compritol® 888 + Labrasol®	77±1	72±6

IV - CONCLUSION

Non-ionic surfactants in Compritol® modify the release of ibuprofen after Hot-Melt Coating.

Surfactants should be miscible with Compritol®, water-soluble and liquid.

Labrasol® is the most promising surfactant as it allows obtaining the greatest DE after compaction and dissolution profiles equivalent whatever the dosage form.

V - REFERENCE LIST

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